

R-Value: Player and Team Production per Dollar of True NBA Expenditure

Model definition, six-season backtest (2018–19 to 2025–26),
and 2026–27 projections

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Abstract

R-Value measures NBA production per dollar of *true* team expenditure — salary grossed up by each team’s luxury-tax burden under the CBA schedule in force that season — normalized so the league average is 100. Its performance core is a calibrated blend of five statistical pillars; its cost core prices every contract against the era-correct tax structure (2017-CBA brackets through 2022–23, the 2023-CBA regime thereafter). Calibrated on six seasons — 2018–19 and 2021–22 through 2025–26, deliberately excluding the two COVID-distorted years — the performance core identifies All-NBA selections better than PIE, raw plus-minus, or estimated net rating, and a fitted title score ranked the actual champion #1, #2, #6, #1, #2, #3. Under leave-one-season-out cross-validation, in which every fitted parameter is refit without the season it is scored on, it holds **recall@15** = 0.789 (vs. 0.711 for the best unfitted rival), $\rho = 0.914$ against team wins, and a mean champion rank of **2.83** of 30 teams. For 2026–27 the model projects Oklahoma City as champion, New York and Houston as #1 seeds, and Detroit as the league’s most cost-efficient team.

1 Key findings

- **The performance core is the best award predictor tested, and it survives cross-validation.** Over six seasons it recovers 82.2% of actual All-NBA selections in its top 15, versus 71.1% for PIE×minutes, 34.4% for raw plus-minus, and 23.3% for estimated net rating. Refit per fold without the season it is scored on, it still recovers 78.9% — the gap between fitted and honest accuracy is three percentage points (Section 3).
- **Extending the calibration window rebalanced the weights.** On three seasons the fit leaned on box production (0.35) and the impact ratings (0.20 offense, 0.20 defense). On six seasons the offensive and defensive impact weights each settle at **0.15**, box drops to 0.25, and the freed weight flows to PIE (0.30) and load-scaled efficiency (0.15): over a longer window, noisy on-court impact ratings cede ground to the steadier all-in-one and efficiency signals (Section 2.5).
- **Rotation depth, not star power, separates champions.** Fitted on six actual champions, the title score weights the eight-man-rotation term nine times heavier than the star term ($a = 0.2$ on the top-two, $b = 1.8$ on the rotation). Star ceiling matters, but the quality of the de facto playoff rotation is what the fit keeps choosing (Section 4).

- **Champions usually are not value teams.** Across the six title winners, value ranks were 16th, 29th, 14th, 13th, **3rd**, and 14th. The 2021–22 Warriors are the extreme: a \$179M payroll plus a \$170M repeater-tax bill bought a championship at a Team R-Value of 49 — the least efficient title in the sample. Only the 2024–25 Thunder won while near the top of the value board, the signature of a rookie-scale core peaking before its extensions kick in.
- **2026–27:** Oklahoma City is a clear title favorite (score 4.49 vs. New York 3.80, Houston 3.63, Denver 3.42) despite projecting as only the West #4 seed on per-minute quality — the league’s best top-two plus its deepest rotation, now paid for in full (\$234M payroll, \$108M tax). Detroit projects as the most R-Valuable team (159) for the second straight season.

2 The model

2.1 Performance pillars

For each player-season, five pillars are computed and standardized. Let $z(\cdot)$ denote a z-score whose mean and deviation are taken over the qualified pool (players with $m \geq 500$ minutes), and let “lg” denote the minutes-weighted league average. With m the player’s minutes:

$$\text{BOX} = \text{PTS} + 0.7 \text{REB} + 1.4 \text{AST} + 2.2 \text{STL} + 2.0 \text{BLK} - 1.4 \text{TOV} \quad (1)$$

$$z_{\text{box}} = \frac{1}{2} z(36 \text{BOX}/m) + \frac{1}{2} z(\text{BOX}) \quad (\text{rate} + \text{volume}) \quad (2)$$

$$z_{\text{eff}} = z\left[(\text{TS}\% - \text{TS}\%^{\text{lg}}) \cdot \text{USG}\%\right] \quad (\text{load-scaled efficiency}) \quad (3)$$

$$z_{\text{off}} = z\left[\left(0.6 \Delta E_{\text{OFF}} + 0.4 \Delta \text{OFF}\right) \cdot \frac{m}{m+500}\right] \quad (4)$$

$$z_{\text{def}} = z\left[\left(0.6 \Delta E_{\text{DEF}} + 0.4 \Delta \text{DEF}\right) \cdot \frac{m}{m+500}\right] \quad (5)$$

$$z_{\text{adv}} = z(\text{PIE}) \quad (\text{game-event share}) \quad (6)$$

where Δ terms are the player’s (estimated) offensive/defensive rating relative to league average, and the $m/(m+500)$ factor is a reliability shrinkage that pulls low-minute impact ratings toward zero.

2.2 Composite score, availability, and WPS

$$\text{CPI} = 0.25 z_{\text{box}} + 0.15 z_{\text{eff}} + 0.15 z_{\text{off}} + 0.15 z_{\text{def}} + 0.30 z_{\text{adv}} \quad (\text{calibrated, Sec. 2.5}) \quad (7)$$

$$\text{PS} = 100 \Phi(\text{CPI}) \quad (\Phi = \text{standard normal CDF}) \quad (8)$$

$$A = \sqrt{\min(m, 2200)/2200} \quad (\text{availability credit}) \quad (9)$$

$$\text{WPS} = \text{PS} \cdot (0.55 + 0.45 A) \quad (\text{weighted performance score}) \quad (10)$$

PS maps CPI onto a 0–100 percentile-like scale; WPS discounts it for games missed, with a floor so that an injured star is dampened, not erased.

2.3 Cost side: pricing each era’s CBA

Team payroll P is the sum of guaranteed salaries (dead money included, two-way contracts excluded). The luxury tax $T(P)$ is computed from the exact bracket schedule of the applicable league

year, using repeater rates where a team qualified:

$$T(P) = \sum_{k \geq 1} r_k \cdot \min(w, \max(0, P - \tau - (k-1)w)), \quad (11)$$

with τ the tax line and w the bracket width. Under the 2017 CBA (through 2022–23): $w = \$5\text{M}$, rates 1.50, 1.75, 2.50, 3.25, ... (+1.00 each for repeaters). Under the 2023 CBA (2025–26 rates, width indexed to cap growth): 1.00, 1.25, 3.50, 4.75, ... standard and 3.00, 3.25, 5.50, 6.75, ... repeater, +0.50 per bracket beyond the fourth. Total expenditure and each player’s effective cost are then

$$\text{TE} = P + T(P), \quad \text{EC}_i = s_i \cdot \frac{\text{TE}}{P}, \quad \text{CI}_i = \text{clip}\left(\frac{\text{EC}_i}{\overline{\text{TE}}}, 0.002, 0.40\right), \quad (12)$$

where s_i is the player’s salary and $\overline{\text{TE}}$ the league-mean total expenditure that season. Every salary dollar on a taxpaying roster is grossed up by that team’s tax multiplier: a \$40M contract on a repeater-taxpayer costs its franchise far more than \$40M.

2.4 R-Value

$$\textbf{Player:} \quad R_i = 100 \cdot \text{normalize} \left[\text{WPS}_i \cdot \left(\frac{\overline{\text{CI}}}{\text{CI}_i} \right)^{0.35} \right] \quad (13)$$

$$\textbf{Team:} \quad \text{TeamR} = 100 \cdot \text{normalize} \left[\frac{\sum_i \text{WPS}_i m_i}{\text{TE}} \right] \quad (14)$$

$$\textbf{Title:} \quad \text{TITLE} = z(\text{PERF}) + a z(\text{TOP2}) + b z(\text{ROT8}), \quad a = 0.2, \quad b = 1.8 \quad (15)$$

where `normalize` divides by the league mean (so 100 is average) and the 0.35 exponent dampens cost leverage so minimum contracts cannot trivially top the list. $\text{PERF} = \sum \text{WPS}_i m_i / \sum m_i$ is minutes-weighted team quality; TOP2 is the mean WPS of the two best players; and ROT8 is the minutes-weighted WPS of the eight most-played players — the de facto rotation that playoff basketball compresses to.

2.5 Calibration and data

The five CPI weights are fitted by a two-stage simplex grid search maximizing an equal blend of All-NBA recall@15 and Spearman correlation with team wins, averaged over the six calibration seasons. The title premiums (a, b) are then fitted on a 0–2.5 grid to minimize the six actual champions’ mean title-score rank, preferring smaller premiums on ties.

Two seasons inside the window are deliberately excluded: 2019–20 (the bubble season; teams played uneven schedules, breaking the availability model) and 2020–21 (72 games; luxury-tax bills were pro-rated at settlement, breaking the cost model). Award eligibility follows each era’s rules: the 65-game requirement applies from 2023–24 only, with a 40-game floor earlier. Pre-2023 salaries are reconstructed from per-season contract archives, individual player salary histories where re-signings purged the archive rows, and hand-verified patches for players absent from all sources; team payroll totals are validated against independent per-team aggregates for all 30 teams in every season.

Table 1: Predicting All-NBA selections: recall@15 among award-eligible players, by season. R-Value uses its cost-free performance core (WPS).

Metric	18–19	21–22	22–23	23–24	24–25	25–26	Mean
R-Value (WPS core)	0.800	0.867	0.733	0.933	0.800	0.800	0.822
PIE $\times$ minutes	0.733	0.733	0.733	0.733	0.600	0.733	0.711
Raw plus-minus	0.333	0.400	0.267	0.333	0.400	0.333	0.344
Estimated net rating	0.200	0.267	0.200	0.200	0.267	0.267	0.233

Table 2: Predicting regular-season wins: Spearman rank correlation, by season.

Metric	18–19	21–22	22–23	23–24	24–25	25–26	Mean
Team net rating	0.976	0.950	0.931	0.955	0.952	0.951	0.952
R-Value perf core	0.935	0.902	0.880	0.924	0.932	0.927	0.917
Team R-Value (per dollar)	0.531	0.482	0.297	0.499	0.633	0.813	0.543
Payroll	0.431	0.360	0.575	0.611	0.129	0.393	0.417

3 Validation against rival metrics

Three observations. First, R-Value beats or ties every rival on All-NBA prediction in five of six seasons (2022–23 is a tie with PIE; voters that year rewarded narrative picks like Fox and Randle that no impact metric selects). Second, team net rating edges the performance core on wins (0.952 vs. 0.917) — but net rating *is* point differential, which is near-definitionally wins; it cannot rank players, price contracts, or be projected forward from individual contracts. R-Value does all three from one framework. Third, payroll alone correlates with wins at only 0.417 — money buys far less than reputation suggests, which is precisely the inefficiency R-Value quantifies.

3.1 Out-of-sample validation

The figures above are *in-sample*: the pillar weights and title premiums were grid-searched to maximize exactly those quantities on exactly those seasons. The rival metrics have no fitted parameters, so this comparison flatters R-Value. To correct for it, every fitted parameter is refit on five seasons and scored on the sixth, rotating through all six.

Table 3: Leave-one-season-out cross-validation. Each row refits the five pillar weights and both title premiums *without* the held-out season, then scores that season.

Held out	Recall @15	ρ wins	Champ rank	Weights chosen					<i>a</i>	<i>b</i>
				box	eff	off	def	PIE		
2018–19	0.800	0.935	1	0.25	0.15	0.15	0.15	0.30	0.2	1.8
2021–22	0.800	0.882	3	0.40	0.05	0.15	0.20	0.20	1.5	2.4
2022–23	0.733	0.880	6	0.25	0.15	0.15	0.15	0.30	0.2	1.8
2023–24	0.800	0.926	1	0.25	0.20	0.15	0.15	0.25	0.1	2.1
2024–25	0.800	0.932	3	0.25	0.15	0.15	0.15	0.30	0.0	1.8
2025–26	0.800	0.927	3	0.25	0.15	0.15	0.15	0.30	0.2	1.8
Mean	0.789	0.914	2.83							

The degradation is small: recall@15 falls from 0.822 to 0.789, still comfortably ahead of

PIE $\times$ minutes at 0.711; the wins correlation falls from 0.917 to 0.914; mean champion rank slips from 2.50 to 2.83. The weights are also stable — four of six folds select an identical vector, the offensive-impact weight is 0.15 in *all six*, and the defensive weight varies by 0.05. Only the 2021–22 fold diverges materially, leaning on box production instead of PIE.

The title premiums tell the same story: the rotation term b lands in $[1.8, 2.4]$ in every fold while the star term a sits at or below 0.2 in five of six. Whatever the precise magnitudes, the fit keeps choosing rotation depth over star ceiling, and it does so without having seen the champion it is asked to rank.

4 Historical backtest: six seasons, six champions

4.1 Did the title score find the champion?

Table 4: Top five title scores per season vs. what actually happened. ★ = actual champion. PO-W = playoff wins (16 = title).

Season	Team	Title	Perf	TOP2	ROT8	Wins	PO-W
2018–19	TOR ★	5.12	68.2	92.5	76.8	58	16
	GSW	4.52	64.8	98.2	74.3	57	14
	MIL	4.38	62.9	94.1	75.1	60	10
	UTA	3.58	64.0	89.7	69.7	50	1
	PHI	3.39	61.3	92.4	69.9	51	7
2021–22	BOS	5.09	67.1	92.9	74.7	51	14
	GSW ★	3.72	61.7	87.5	70.0	53	16
	PHX	3.72	62.6	92.2	69.0	64	7
	UTA	3.16	59.2	92.3	67.6	49	2
	MEM	2.99	60.3	85.2	66.5	56	6
2022–23	BOS	4.45	65.4	90.4	71.0	57	11
	CLE	4.02	63.6	89.6	69.7	51	1
	SAC	2.81	59.7	90.4	65.1	48	3
	MIL	2.67	56.5	92.6	66.1	58	1
	NYK	2.40	56.7	84.5	65.2	47	6
	<i>champion DEN ranked #6 (coasted to perf #9; playoff Jokić broke the model)</i>						
2023–24	BOS ★	5.29	70.7	91.6	78.8	64	16
	MIN	3.98	68.5	90.1	71.1	56	9
	OKC	3.28	61.5	94.9	70.2	57	6
	NOP	2.92	61.3	83.8	68.9	49	0
	LAC	2.80	60.0	91.0	68.1	51	2
2024–25	CLE	4.89	65.3	94.5	78.9	64	5
	OKC ★	4.32	67.1	94.5	73.5	68	16
	BOS	4.31	68.0	88.0	73.6	61	6
	MIN	3.20	63.3	86.1	68.4	49	9
	NYK	2.84	60.9	89.7	66.8	51	10
2025–26	SAS	4.06	66.0	89.0	72.6	62	13
	BOS	3.55	65.0	88.0	69.6	56	3
	NYK ★	3.37	61.5	89.6	70.3	53	16
	OKC	3.18	62.0	97.1	67.7	64	11
	MIN	3.12	60.4	85.4	69.6	49	6

The champion ranked **#1, #2, #6, #1, #2, #3** — mean rank 2.50 out of 30 teams, with four of six champions in the top two. The one real miss is instructive: the 2022–23 Nuggets ranked ninth in regular-season performance because Jokić’s supporting cast coasted, then flipped a switch in April — exactly the failure mode a regular-season metric cannot see. The 2021–22 season shows the

opposite face: Boston graded far ahead of the field, reached the Finals, and lost it to the Warriors in six — a near-hit ranked second.

4.2 Value and winning are different axes

Table 5: Most R-Valuable teams per season vs. where the champion ranked in value terms (of 30).

Season	Top three by Team R-Value	Champion (TeamR)	Champ. value rank
2018–19	PHI 139, UTA 138, IND 136	TOR (105)	16th
2021–22	MEM 152, CHA 149, PHX 144	GSW (49)	29th
2022–23	MEM 151, SAC 142, NYK 141	DEN (107)	14th
2023–24	OKC 143, ORL 142, IND 139	BOS (109)	13th
2024–25	DET 153, CLE 151, OKC 151	OKC (151)	3rd
2025–26	MIA 141, SAS 138, DET 136	NYK (103)	14th

The pattern across eight years is stable: championships are bought at roughly league-average efficiency or worse, and the value leaderboard is populated by young teams on rookie-scale contracts — some of which (OKC 2023–24 → champion 2024–25) convert value into titles, and some of which (MEM 2021–23) never do. The 2021–22 Warriors bookend the distribution: with a \$179M payroll and a \$170M repeater-tax bill, their Team R-Value of 49 is the least efficient championship in the sample — and a working definition of what “buying a title” looks like in R-Value terms.

4.3 All-NBA through the R-Value lens

Of the 90 actual All-NBA selections across the six seasons, the performance core placed **74 (82.2%)** in its top fifteen eligible players (12, 13, 11, 14, 12, and 12 per season). The same tables expose the price of stardom in every era: in 2021–22, rookie-scale Luka Dončić (WPS 96.9, \$10.2M, R = 210) and Trae Young (WPS 95.1, \$8.3M, R = 221) delivered first-five production at a fifth of the veteran max; in 2025–26 the same slot belongs to Jalen Duren (WPS 95.7, \$6.5M, R = 253) and Victor Wembanyama (WPS 96.2, \$13.4M, R = 197).

5 2026–27 projections

Player CPI is projected as a 0.55/0.30/0.15 blend of the last three seasons plus an age-curve adjustment; minutes from a two-season blend, rescaled to a full 19,680-minute roster; rosters and salaries reflect contracts signed as of July 5, 2026 (431 players, of whom 41 rookies/unknowns enter at replacement level). Unsigned free agents are absent by construction.

5.1 Predicted standings

5.2 Title picture

OKC’s projected eight-man rotation is more than five WPS points clear of every other contender — the same profile the backtest ranked top-two in 2024–25 — while Denver enters the contender group on the strength of the Jokić–Murray top-two (92.9). The champion pick is robust to the calibration window: both the three-season and six-season fits select OKC, but the six-season fit

Table 6: Projected 2026–27 standings by performance core. TeamR = projected Team R-Value; payroll and tax in \$M.

East							West						
#	Team	Perf	Title	TeamR	Pay	Tax	#	Team	Perf	Title	TeamR	Pay	Tax
1	NYK	60.9	3.80	95	215	46	1	HOU	60.5	3.63	125	199	0
2	CLE	55.4	2.40	123	184	0	2	DEN	59.2	3.42	83	220	73
3	BOS	55.3	2.47	109	203	5	3	SAS	58.9	3.00	122	198	0
4	TOR	55.0	2.29	111	202	1	4	OKC	58.0	4.49	69	234	108
5	DET	54.4	2.67	159	140	0	5	MIN	56.5	2.22	80	219	70
6	ORL	54.4	1.69	80	224	55	6	LAL	52.8	1.46	112	192	0
7	PHI	51.6	0.86	103	203	2	7	GSW	52.7	0.90	117	184	0
8	IND	51.3	0.54	98	208	7	8	POR	46.9	−1.17	101	191	0
9	MIA	49.5	0.18	102	199	0	9	PHX	45.6	−0.87	73	214	43
10	ATL	48.4	0.60	104	191	0	10	DAL	43.9	−1.48	101	178	0
11	CHA	47.2	−0.01	115	169	0	11	NOP	42.6	−1.99	91	192	0
12	MIL	44.0	−0.40	108	167	0	12	LAC	40.6	−3.00	122	136	0
13	CHI	41.7	−2.14	104	165	0	13	UTA	37.6	−4.37	88	175	0
14	BKN	36.2	−4.38	90	164	0	14	SAC	37.1	−4.56	60	219	35
15	WAS	32.5	−6.76	73	183	0	15	MEM	33.1	−5.49	86	158	0

Table 7: Top five 2026–27 title scores. OKC wins on rotation depth despite projecting fourth in the West on per-minute quality.

Rank	Team	Conf. seed	Title score	Perf	TOP2	ROT8
1	OKC	West #4	4.49	58.0	93.6	71.6
2	NYK	East #1	3.80	60.9	88.6	66.2
3	HOU	West #1	3.63	60.5	87.1	65.7
4	DEN	West #2	3.42	59.2	92.9	64.8
5	SAS	West #3	3.00	58.9	85.9	63.5

widens the margin. The cost side is equally striking: OKC’s Team R-Value falls to 69 (\$234M payroll, \$108M tax), completing the conversion from the league’s best value team (2023–24) to its biggest spender — the exact path the 2021–22 Warriors took to their inefficient title.

5.3 Predicted All-NBA

5.4 Where the value is

Detroit is the standout: a projected playoff seed on a \$140M, tax-free payroll, anchored by two projected All-NBA players of whom one (Duren) earns \$9M. The six-season backtest gives that profile a precedent with both outcomes: OKC 2023–24 converted the same shape into a title within two seasons; Memphis 2021–23 topped the value board twice and never escaped the second round.

6 Limitations

- **COVID seasons excluded.** 2019–20 and 2020–21 are omitted because uneven schedules and pro-rated tax bills break the availability and cost models respectively; the calibration window is therefore six seasons spanning eight years, not eight consecutive seasons.
- **Pre-2023 salary reconstruction.** Older seasons are rebuilt from contract archives that

Table 8: Projected 2026–27 All-NBA: top 15 by projected WPS among players projected for ≥ 1600 minutes. Salary in \$M.

Team	Player	Club	Age	WPS	Salary	R-Value
1st	Shai Gilgeous-Alexander	OKC	28	99.8	40.8	126
1st	Nikola Jokić	DEN	32	99.8	59.0	114
1st	Luka Dončić	LAL	28	96.9	51.0	129
1st	Victor Wembanyama	SAS	23	94.0	16.9	184
1st	Jalen Duren	DET	23	92.9	9.0	227
2nd	Donovan Mitchell	CLE	30	91.9	50.1	123
2nd	Karl-Anthony Towns	NYK	31	91.5	57.1	109
2nd	Anthony Edwards	MIN	25	90.7	48.9	111
2nd	Cade Cunningham	DET	25	90.5	50.1	121
2nd	Tyrese Haliburton	IND	27	90.5	48.9	121
3rd	Kawhi Leonard	TOR	36	88.9	50.3	119
3rd	Kevin Durant	HOU	38	87.8	43.9	123
3rd	Chet Holmgren	OKC	25	87.3	41.2	110
3rd	Evan Mobley	CLE	26	87.1	50.1	117
3rd	Jarrett Allen	CLE	29	86.7	28.0	142

Table 9: Projected most R-Valuable teams (left) and highest player R-Values among players projected for ≥ 1200 minutes (right, top 8).

#	Team	TeamR	Proj. seed	#	Player	R	WPS
1	DET	159.0	East #5	1	Neemias Queta (BOS)	282	76.0
2	HOU	124.7	West #1	2	Moussa Diabaté (CHA)	245	63.6
3	CLE	123.1	East #2	3	Ryan Kalkbrenner (CHA)	237	61.1
4	LAC	121.9	West #12	4	Jalen Duren (DET)	227	92.9
5	SAS	121.7	West #3	5	Kel’el Ware (MIL)	219	71.0
6	GSW	117.0	West #7	6	Ryan Rollins (MIL)	207	63.6
7	CHA	114.6	East #11	7	Payton Pritchard (BOS)	200	78.2
8	LAL	112.3	West #6	8	Cam Spencer (MEM)	197	50.7

purge superseded deals; coverage is restored via individual salary histories and hand patches, and team totals are validated against independent aggregates, but a handful of departed role players carry imputed minimum salaries.

- **Roster incompleteness for 2026–27.** Projections use contracts signed as of the run date; unsigned free agents (e.g., LeBron James, James Harden) and future signings are absent, which depresses their destination teams. 2026–27 cap, tax, and apron figures are league projections.
- **Six champions is still a small fit.** The title premiums $(a, b) = (0.2, 1.8)$ carry wide uncertainty. Cross-validation (Section 3.1) shows the *direction* is stable — rotation over star in all six folds — but b ranges from 1.8 to 2.4 across them, and mean champion rank is the weakest of the three validated quantities.
- **Descriptive tables use the full-sample fit.** The per-season tables in Section 4 and the 2026–27 projections are generated with weights fit on all six seasons, which is the right choice for description and forecasting. Only Section 3.1 reports accuracy the model had not already

seen; cite those figures, not the in-sample ones.

Generated from the R-Value pipeline (`backtest.py`, `predict.py`); data: NBA Stats API and public salary records, seasons 2018–19 through 2025–26 (ex-COVID), contracts as of July 5, 2026. League average = 100 for all R-Values.